

Week 3 - Day 1: Machine Learning Fundamentals

5-Minute Review Notes

Day Goal: Understand the basic language and workflow of Machine Learning before moving into deeper algorithms: features, target, supervised vs unsupervised learning, regression vs classification, train/test splitting, model training, prediction, and evaluation.

1. From Data Analysis to Machine Learning

Week 2 focused on understanding existing data: **Load -> Inspect -> Clean -> Analyze -> Visualize -> Interpret**. Machine Learning extends that workflow by using patterns in historical data to make predictions or discover structure.

ML workflow: Load -> Inspect -> Clean -> Analyze -> Prepare -> Train Model -> Predict -> Evaluate

Example: In the Canadian Housing Analysis project, EDA can show that larger homes tend to have higher prices. Machine Learning asks the next question: **Can a model learn from existing houses and predict the price of a new house?**

2. Traditional Programming vs Machine Learning

Traditional Programming	Machine Learning
Data + human-written rules -> output	Features + known answers -> algorithm -> trained model
Programmer defines the logic.	Algorithm learns patterns/parameters from data.
Example: if score >= 50 -> Pass	Example: learn how study hours relate to exam score

3. Features (X) and Target (y)

Features (X) are the input information used by the model. **Target (y)** is the answer/value we want the model to predict.

Mental model: X -> Model -> y

Housing example:

X: Area_sqft, Bedrooms, Year_Built

y: Price

In Pandas, multiple feature columns are commonly selected as a DataFrame, while a single target column is commonly a Series.

```
x = df[["Area_sqft", "Bedrooms", "Year_Built"]]
y = df["Price"]
```

4. Supervised vs Unsupervised Learning

Type	What is available?	Goal
Supervised Learning	X + known y	Learn the relationship and predict y for new data
Unsupervised Learning	X only; no predefined y	Discover hidden structure, patterns, or groups

Important Day 1 correction: In an unsupervised customer-clustering problem, the discovered groups (Cluster 1, Cluster 2, etc.) are outputs discovered by the algorithm. They are not a known target y supplied before learning.

5. Regression vs Classification

Problem	Prediction	Examples
Regression	A numerical value	House price, salary, temperature, sales
Classification	A class/category	Spam/Not Spam, Approved/Rejected, Cat/Dog/Horse

Key distinction: Classification is determined by the prediction goal, not simply by whether a column is stored as Boolean, text, or numbers. Approved/Rejected, True/False, and 1/0 can all represent classes.

6. What Is a Model?

A model is the learned mathematical relationship or pattern created from training data. The algorithm uses examples to find suitable parameters; we do not manually write the final prediction rule.

Simple conceptual example:
 $\text{Price} \approx 300 \times \text{Area} + 150000$

The important point is not this exact formula. The important point is that **training learns the parameters from data.**

7. Training Data vs Testing Data

A model must be evaluated on examples it did not use for learning. Therefore, a dataset is commonly split into a training set and a testing set.

Dataset Part	Purpose	Common Variables
Training data	Teach the model	X_train, y_train
Testing data	Unseen exam for the trained model	X_test, y_test
Prediction	Model's answers for X_test	y_pred

Useful analogy: Training data is the study/practice material. Testing data is the unseen exam. A high score on material already memorized does not prove the model can generalize.

8. Core scikit-learn Workflow

Day 1 used Linear Regression from scikit-learn to predict exam score from study hours.

```
X = df[["Study_Hours"]]
y = df["Exam_Score"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Why define X and y separately? It makes the ML workflow easier to read and maintain, especially when X later contains many feature columns.

9. fit() vs predict()

Method	Meaning	Uses
.fit()	LEARN	X_train + y_train
.predict()	PREDICT using what was learned	X_test

During prediction, the model receives **X_test**, not y_test. After predictions are produced, **we compare y_pred with y_test** to judge model performance.

10. Day 1 Practical Exercise - Study Hours -> Exam Score

Dataset used:

Study_Hours = 1 to 10

Exam_Score = 45, 50, 55, 60, 65, 70, 74, 80, 86, 92

The model was trained using an 80/20 train-test split with random_state=42. The two held-out examples produced the following predictions:

Actual	Predicted	Approx. Absolute Error
86	85.577586	0.42
50	49.672414	0.33

Interpretation: ML predictions do not need to exactly equal the actual values. The next important idea is measuring how wrong predictions are - model evaluation.

11. Scenario Assessment - Correct Answers

Scenario	X (Features)	y (Target)	Learning Type	Problem
Employee salary	Experience, Education, Job Role	Salary	Supervised	Regression
Loan approval	Income, Credit Score, Debt	Loan Approved	Supervised	Classification
Customer grouping	Age, Income, Annual Spending	None	Unsupervised	Clustering

12. Day 1 Assessment Questions - Proper Answers

Q1. Traditional programming vs Machine Learning?

Traditional programming uses data plus rules written by the programmer to produce an output. Machine Learning uses data and examples to learn patterns/parameters, creating a model that can make predictions on new data.

Q2. Feature vs target?

Features (X) are the input information used to make a prediction. The target (y) is the value or class we want to predict.

Q3. Why is house-price prediction regression?

Because the target is a numerical quantity such as \$650,000 or \$825,500.

Q4. Why is spam/not-spam classification?

Because the model predicts which category/class an email belongs to. Classification can have two classes or multiple classes.

Q5. Why separate training and testing data?

The model learns from training data and is then evaluated on unseen testing data. This helps measure whether it can generalize rather than simply perform well on examples it already saw.

Q6. What does `model.fit(X_train, y_train)` do?

It trains the model by learning the relationship between the training features and their known target values.

Q7. What does `model.predict(X_test)` do?

It uses the trained model to generate predicted target values for the unseen test features.

Q8. Why not give `y_test` during prediction?

`y_test` contains the actual answers. The model should predict from `X_test` alone; afterward, `y_pred` is compared with `y_test` to evaluate performance.

13. Quick Review Cheat Sheet

Term	Remember
X	Features / inputs
y	Target / answer to predict
Supervised	X + known y
Unsupervised	X only; discover patterns
Regression	Predict a numerical value
Classification	Predict a category/class
Clustering	Discover similar groups without a predefined target
<code>train_test_split()</code>	Separate learning data from unseen evaluation data
<code>random_state=42</code>	Makes the random split reproducible
<code>.fit()</code>	Learn
<code>.predict()</code>	Predict
<code>y_pred</code>	Model's predictions
<code>y_test</code>	Actual held-out answers used for evaluation

14. Day 1 Completion Summary

Week 3 - Day 1: COMPLETE

Completed concepts:

- Machine Learning fundamentals
- Traditional programming vs ML
- Features (X) and target (y)
- Supervised vs unsupervised learning
- Regression vs classification
- Clustering concept
- Training vs testing data
- `train_test_split()`
- LinearRegression model creation
- `model.fit()` and `model.predict()`
- `y_test` vs `y_pred`
- First end-to-end ML prediction exercise
- Conceptual correction: unsupervised clustering has no predefined y

Progress: Week 1 Python - 7/7 complete | Week 2 Data Analysis - 7/7 complete | Week 3 Machine Learning - Day 1 complete